

Raging Against the Machine: Assessing the Impact of Regulations in Algorithmically-Priced Markets

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The views herein are those of the authors and do not represent those of the Deutsche Bundesbank or the Eurosystem.

Motivation

Market design is algorithm design

- Prices are increasingly set by algorithms that adapt to what has been profitable, e.g., in financial markets and online retail.
- Such markets admit many self-confirming price patterns: cycles, focal prices, everything between. A rule shifts which one is learned.

Regulating a market of algorithms

Imposing rules in an algorithmic market does not just constrain prices; it rewrites the problem the algorithms are solving. The outcome is whatever they then learn, which need not be what the rule intended.

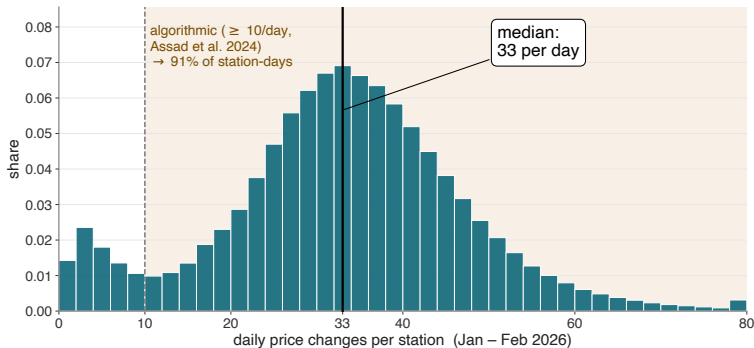
The German fuel market as an example

- On 1 April 2026, the German government introduced a new law limiting fuel retailers to increase prices once per day, at noon.
- Price decreases are freely permitted throughout the day; violations of the rule may be subject to a fine of up to €100,000.
- Early evidence: margins rose by 5–6 cents per litre (Jung et al., 2026).

Question

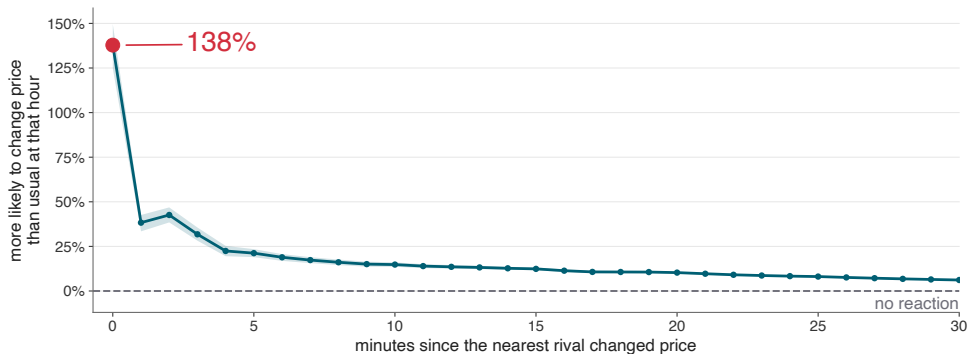
- Why did a rule aimed at limiting frequent price changes end up increasing margins?

The rule landed in a market populated by algorithmic-price setters...



- The median station reprices **33 times a day** — well above the threshold previously used (≥ 10 times per day) to identify algorithmic pricers (Assad et al., 2024).

...who respond rapidly when a rival changes prices



- Stations are 138% more likely to change prices the minute a rival does than they usually do.
- The pattern extends throughout the day and overnight. [\[details\]](#)

A price rule becomes a learning rule

- On paper, the reform regulates **posted prices**: firms may raise prices only at noon.
- In an algorithmic-priced market, the same rule changes the **action set** and its **payoffs**.
- **Our claim**: the rule changed which equilibria pricing algorithms learn to coordinate on.
- **Roadmap**: Mechanism $\rightarrow$ Data and evidence $\rightarrow$ Takeaways. [related literature]

Mechanism



Canonical model: alternating price competition

- Based on Maskin and Tirole (1988): two firms, perfect substitutes, zero costs, discrete time.
- Demand is $D(p) = 1 - p$. Instantaneous profit of firm i is

$$\pi_i(p_i, p_j) = \begin{cases} p_i (1 - p_i), & p_i < p_j \\ \frac{1}{2} p_i (1 - p_i), & p_i = p_j \\ 0, & p_i > p_j. \end{cases}$$

- Firms alternate in setting prices. Each price is held for two periods.
- Markov strategies: a firm conditions only on the rival's current price.

The game has two kinds of Markov-perfect equilibria

- **Focal price:** both firms hold a common price. Undercutting is met with permanent retaliation, so deviation is not profitable.
- **Edgeworth cycle:** firms undercut until price nears cost; one firm then resets high, loses share for a period, and the descent restarts.

The theory result gives multiplicity, not selection

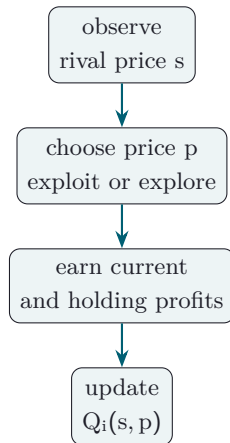
Both types of equilibrium exist. The open question is which one a learning process selects.

Q-learning in simple terms

- Each firm keeps a table of values $Q_i(s, p)$.
- Each entry captures, “how profitable has price p been when the rival’s current price is s ?”
- The firm mostly chooses the best-valued price, but sometimes experiments.
- After observing profits, it updates only the cell it actually used.

Q-learning and equilibrium selection

Q-learning does not solve for an equilibrium; it reinforces profitable strategies and can settle in different self-confirming price patterns.



The Q-update: reinforcement with a continuation value

- Prices lie on the grid $\Omega \equiv \{0, \frac{1}{24}, \frac{2}{24}, \dots, 1\}$.
- In step t and state s , firm i chooses

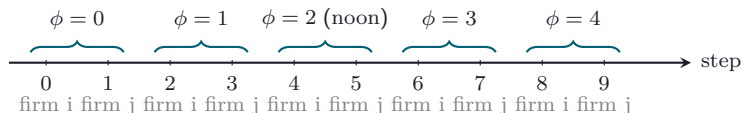
$$p = \begin{cases} \arg \max_{p'} Q_i(s, p'), & \text{with prob. } 1 - \varepsilon_t, \\ \text{a random price in } \Omega, & \text{with prob. } \varepsilon_t. \end{cases}$$

- In step $t + 2$, after the rival has moved, firm i updates

$$Q_i(s, p) \leftarrow (1 - \alpha)Q_i(s, p) + \alpha \left[\underbrace{r_i^{\text{set}}}_{\text{reward in step } t} + \delta \underbrace{r_i^{\text{hold}}}_{\text{reward in step } t+1} + \delta^2 \max_{p'} Q_i(s', p') \right].$$

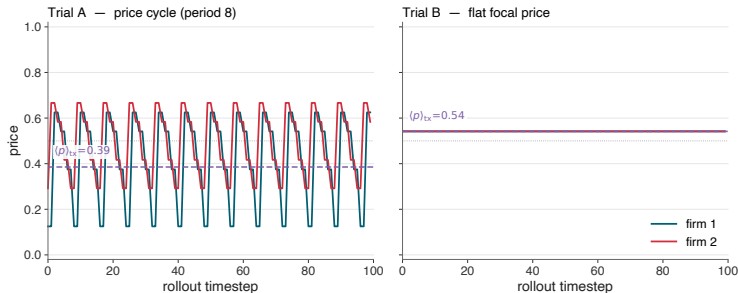
Simulating a duopoly: where noon enters

- **The day.** Two firms alternate; five phases a day, one of them **noon**.



- **A trial.** Random initial Q-values and prices; 2×10^6 learning steps; $\alpha = 0.3$, $\delta = 0.9$, $\varepsilon_t = 2^{-t/H}$, half-life $H = 10^5$.
- **The experiment.** Run **1000** independent trials that differ only in initial values, initial prices, and exploration draws.

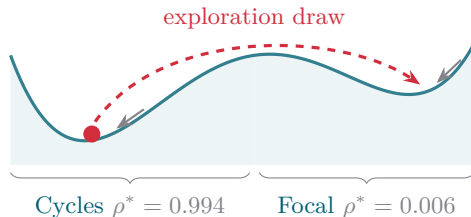
Learning selects among the game's equilibria



- Same game, same parameters; different learning histories can converge to different price patterns.
- Edgeworth cycles in 996 trials and focal prices in 4 trials.
- We summarize profitability using

$$CI = \frac{\bar{\pi} - \pi_{\text{comp}}}{\pi_{\text{mono}} - \pi_{\text{comp}}}, \quad CI_A = 0.47, \quad CI_B = 0.98.$$

Formalizing selection: a master equation over attractors



- Q-learning is a **deterministic map**: initial condition $\rightarrow$ attractor.
- Exploration creates rare jumps between attractors; the transition rates form $\mathbb{W}$. [\[details\]](#)
- The long-run selection distribution, ρ , satisfies

$$\dot{\rho} = \mathbb{W}^\top \rho \quad \implies \quad \mathbb{W}^\top \rho^* = 0.$$

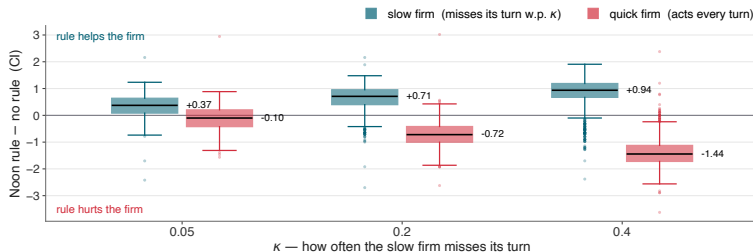
The noon rule shifts learning toward higher-price equilibria

- **Hard rule.** Price increases are allowed only at noon; off-noon, firms may only stay or cut.



- **Only day-long cycles survive:** the daily row holds 82% of trials, up from 26%.
- **More trials stop cycling:** focal-price equilibria hold 12%, up from 0.6%.
- **Peaks move up:** the average top-of-cycle price rises from 0.68 to 0.83.

Noon creates a common reset clock



- The rule hands firms a **fixed moment at which prices may rise**.
- A **slow firm** — misses its turn with probability κ — cannot win the undercutting race, so the common clock is pure gain; a **quick firm** loses the flexibility it was exploiting.

Key mechanism

The regulation does not simply reduce the number of price changes. It changes coordination possibilities within the learning environment.

Three empirical implications

- 1 First stage: price increases should concentrate at noon. [\[details\]](#)
- 2 Outcome: margins should rise after the rule. [\[details\]](#)
- 3 Mechanism: within a local market, slower repricers should gain relative to faster repricers.

Identification logic

The first two patterns can each have alternative explanations. The third is the mechanism's distinctive incidence prediction.

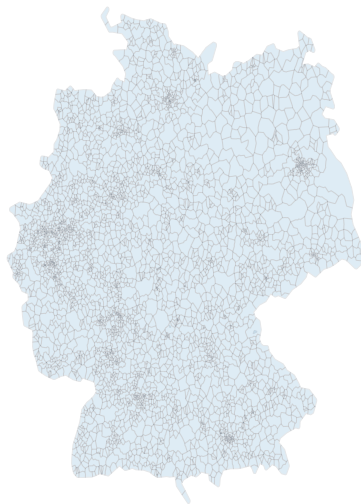
Data and evidence



- **Experiment.** The noon rule takes effect **1 April 2026**, announced three weeks earlier — so vendors reprogrammed rather than retrained.
- **Data.** Tankerkönig prices at 1-minute resolution, wholesale from NYMEX RBOB futures; **13,563** stations, $\geq 95\%$ coverage, E10 throughout.
 - ▶ **Sample period.** 1 Jan – 13 Apr 2026; RDD over 2 Mar – 13 Apr (30 days pre, 13 post).
- **Pricing responsiveness.** Measured before the reform by **how often a station changes price** (median 32 a day) — the empirical counterpart of κ .
- **Margin**, net of wholesale and tax by construction, read at **1pm** and at **end of day**:

$$m_{it} = \frac{p_{it}^{\text{retail}}}{1 + \text{VAT}} - p_t^{\text{wholesale}} - T_t.$$

- Following [Lemus and Luco \(2021\)](#), there are [4,164](#) local markets across Germany.
- Median [3 stations](#) per market (interquartile range 2–5).
- [11%](#) are singletons — one isolated station.
- Median market area $\approx 47 \text{ km}^2$.
- Mostly local-road markets; cities fragment into many small markets.



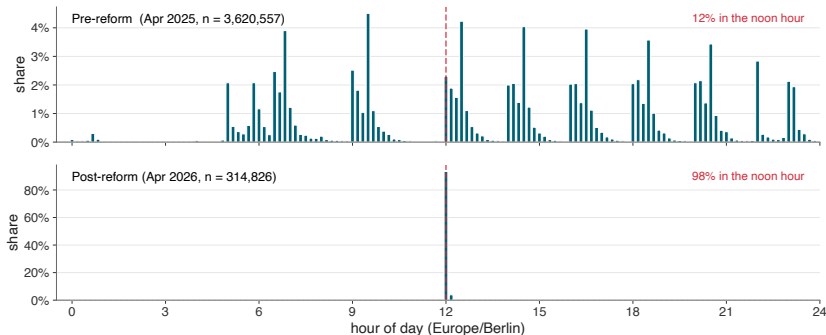
Empirical design: one cutoff, three tests

- **Design.** Time-series RDD — time is the running variable with cutoff on April 1st.

Implication	Test	Fixed effects	SE clustered on
First stage	post-noon – pre-noon spread	station FE	station
Outcome	margin at 1pm and end of day	market FE	local market
Mechanism	margin jump on pre-reform repricing frequency	market $\times$ day FE	local market

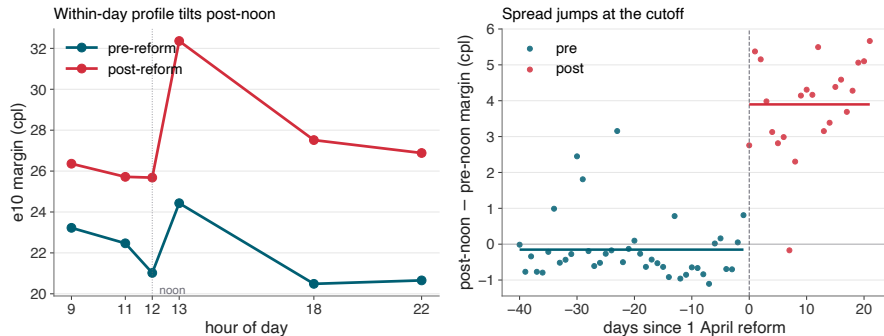
- Autobahn stations are dropped throughout: captive-demand monopolists, not firms on a competitive cycle.

First stage: did the rule bite?



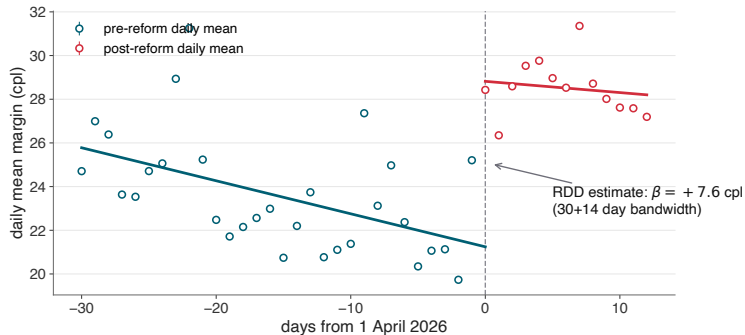
- 0.4% station-days carry a violation (increase prices after 1pm post-reform) of the noon rule
→ compliance is near-total. [\[details\]](#)

First stage: the day is reorganized around noon



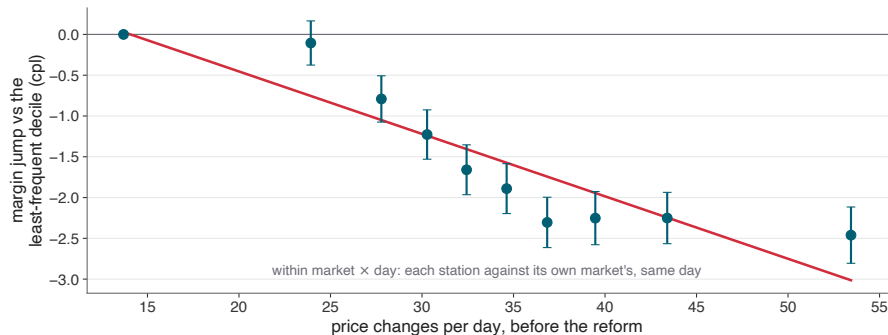
- The intraday profile **tilts**: the post-noon margin rises relative to pre-noon by +6.1 cpl.

Outcome: margins rose sharply after the rule



- 1pm margin +10.9 cpl. The series plotted above is the end-of-day margin, +7.7.
- Net of an Easter placebo against 2023–25 controls, the jump is still +6 to +8 cpl.

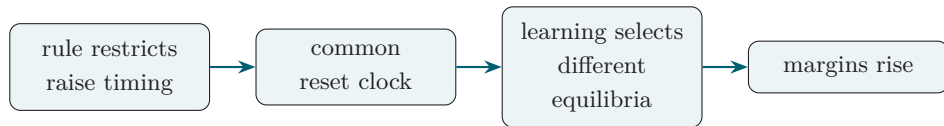
Mechanism: the faster repricers gained less



- Each station is compared with stations in its own market on the same day.
- The busiest decile gained 2.5 cpl less than the quietest; the pattern is monotone across deciles.

Takeaways and policy design

What we have shown

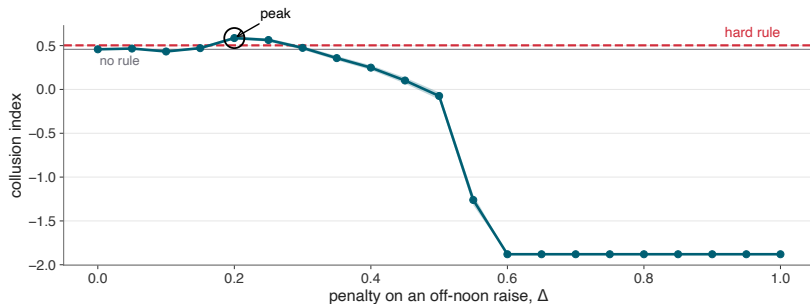


- In the model, the noon rule shifts learning toward daily cycles and focal prices.
- In the data, prices reorganize around noon, margins rise, and fast price setters gain less than slow ones.

Main lesson

A rule written about prices becomes a rule about what algorithms learn.

Rule design matters: hard and soft rules are not the same



- **Hard rule:** an off-noon raise is not allowed. **Soft rule:** it is allowed but fined.

- Hard rules edit the **action set**; soft rules edit the **rewards**.

⇒ CI is **non-monotone** in the fine: moderate fines may sustain coordination, while punitive fines can break it.

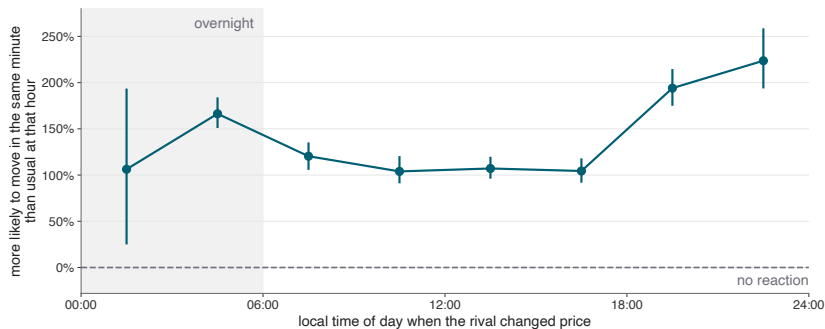
- 1 Regulatory architecture shapes equilibrium selection. A rule can change not just feasible actions, but the set of outcomes that learning settles on.
- 2 Penalties are a lever, but not a monotone one. Moderate fines can sustain collusion; only large ones push the system over the cliff.
- 3 Market design is algorithm design. While inadvertent in the case of the German fuel price rule, it is a potential policy tool that regulators can use to influence algorithmic outcomes.

Thank you!

For questions or comments:

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Backup

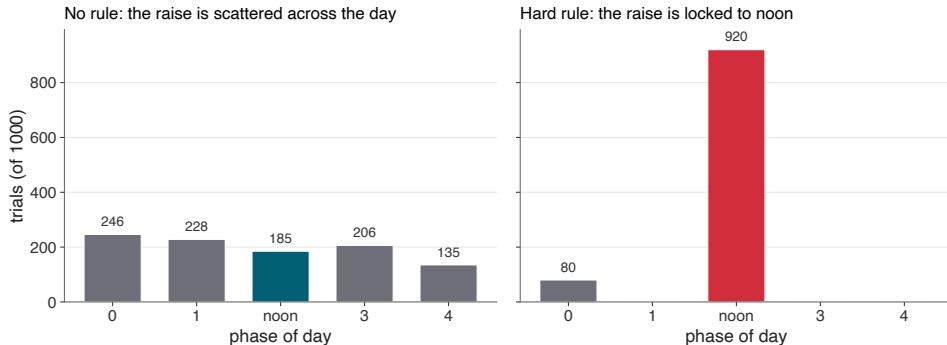


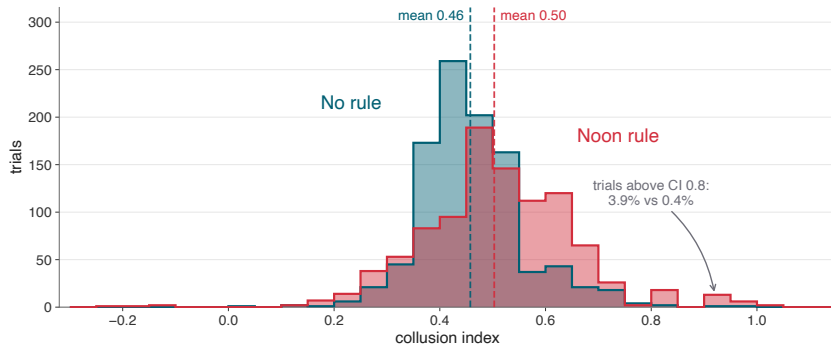
- Never below +100%; 03:00–06:00 (+166%) beats the midday bins (+104% to +107%).
- Raw response rates do fall overnight — but the baseline falls by more.

- Algorithmic collusion, simulated. [Calvano et al. \(2020\)](#); [Klein \(2021\)](#) — Q-learners reach supracompetitive prices.
We ask which of the many equilibria they select.
- Algorithmic pricing, measured. [Assad et al. \(2024\)](#) — adoption raised margins in this market; [Brown and MacKay \(2023\)](#) — pricing speed does.
We study a rule imposed on an already-algorithmic market.
- Edgeworth cycles. [Maskin and Tirole \(1988\)](#); [Byrne and de Roos \(2019\)](#) — multiplicity and coordination in retail fuel.
The multiplicity is theirs; the selection is ours.

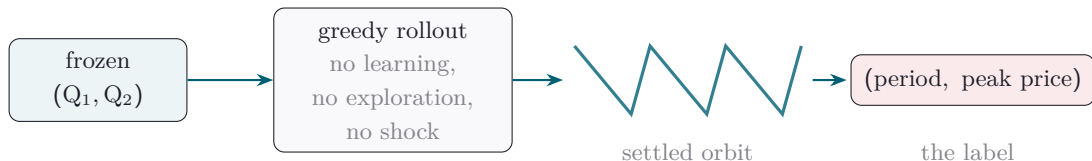
Model backup: bunching at noon is mechanical compliance

[return]



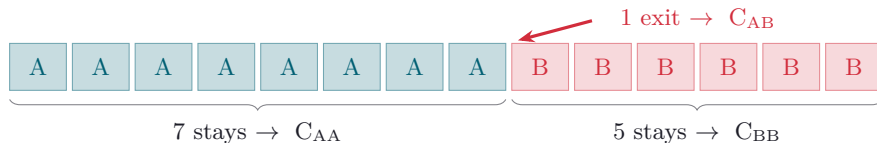


- From the master equation, ρ^* shifts toward daily cycles and focal prices.
- Under focal price, there is no undercutting $\rightarrow$ margins rise.
- Under daily cycles, undercutting does not go as deep $\rightarrow$ the floor rises.



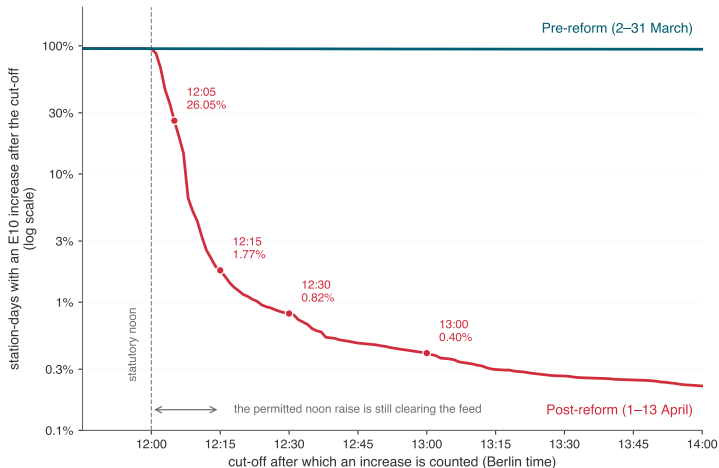
- Freeze the Q-tables and roll the policy forward deterministically: each firm plays argmax against the rival's price, subject to the noon rule. The path settles on an orbit, and its period and peak price are the label.

- After training ends, we **continue** the same Q-tables for 2×10^6 steps at ε **fixed** at 0.001, reading the label every $\Delta t = 4000$ steps — **500 snapshots** $\times$ **1000 seeds**.



- Count **every adjacent pair, stays included**; the row sum of C is the **time spent** in that state.

$$P_{ij} = C_{ij} / \sum_{j'} C_{ij'}, \quad \text{escape rate} = 1 - P_{ii}, \quad W = (P - I) / \Delta t.$$



- A local market is a set of stations a driver would realistically substitute between. We partition Germany's stations bottom-up, following [Lemus and Luco \(2021\)](#):
 - 1 Driving time. Compute pairwise station-to-station driving times by routing on the road network (OSRM).
 - 2 Cluster. Hierarchical agglomerative clustering, average linkage.
 - 3 Cut. Prune the tree at the [90th-percentile inconsistency coefficient](#); singletons have no rival within a 60-minute drive.
- Each market is drawn as the [union of its stations' Voronoi cells](#).

Commercial pricing software is a **two-layer stack**.

Layer 1 — Decision

What should the price be?

- Inputs: rival prices, wholesale cost, local demand signals.
- AI/ML optimization; rule-based matching; manual.

Layer 2 — Execution & compliance

How does it reach pump and regulator?

- Ingest rival feeds; write price back to pump, sign, POS.
- Report to the Market Transparency Unit within **5 minutes**.

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